

Jorge Novoa Contreras

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RESEARCH INTERESTS

Nonlinear analysis, differential equations, periodic boundary value problems, topological degree and continuation methods, variational methods, and monotone operator theory.

EDUCATION

Universidad de Chile

Santiago, Chile

M.Sc. in Applied Mathematics

2025 – 2026

Final grade: **7.0/7.0** (highest possible grade)

Thesis: *Periodic Solutions for Nonlinear Differential Equations in Banach Spaces with Generalized Variable-Exponent Operators*

Developed existence theorems for nonlinear differential equations in Banach spaces. Supervised by **Prof. Raúl Manásevich**.

Thesis Committee: Prof. Jean Mawhin, Prof. Gabrielle Nornberg, and Prof. Alexander Quaas.

Mathematical Engineering

2020 – 2025

Final grade: **7.0/7.0** (highest possible grade)

B.Sc. in Engineering Sciences, major in Mathematics

2020 – 2024

PUBLICATIONS

1. Raúl Manásevich and **Jorge Novoa**, *Global continuum of solutions for systems of ODEs with periodic boundary conditions and generalized variable-exponent operators*, *Communications on Pure and Applied Analysis* **30** (2026), 151–166.

PREPRINTS

1. **Jorge Novoa**, *An equivalence between Schauder's fixed point theorem and periodic solutions in Banach spaces*, preprint, 2026. Submitted for publication. DOI: 10.21203/rs.3.rs-10275062/v1.

RESEARCH EXPERIENCE

Pontificia Universidad Católica de Chile

Santiago, Chile

Research Assistant, FONDECYT Grant 1250522

Sep. 2026 – present

- Conducting research on quasilinear differential equations under the supervision of **Prof. Marta García-Huidobro**.

Universidad de Chile

Santiago, Chile

Research Intern

Dec. 2024 – Feb. 2025

- Conducted research on nonlinear differential equations, focusing on existence results for periodic systems involving ϕ -Laplacian-type and generalized variable-exponent operators, under the supervision of **Prof. Raúl Manásevich**.

TEACHING EXPERIENCE

Universidad de Chile

Santiago, Chile

Assistant Instructor

2022 – 2026

- Conducted problem-solving sessions and tutoring in *Nonlinear Analysis*, *Ordinary Differential Equations*, *Multivariable Calculus*, and *Laboratory: Enhancing Your Mathematical Skills*.

Teaching Assistant

2022 – 2025

- Graded quizzes, tests, and examinations in *Multivariable Calculus*, *Differential and Integral Calculus*, *Complex Variables and Special Functions*, *Linear Algebra*, and *Advanced Calculus and Applications*.

ADDITIONAL EXPERIENCE

Center for Mathematical Modeling (CMM)	Santiago, Chile
Selection Reviewer, International Mathematical Modeling Challenge (IMMC)	2023
• Reviewed mathematical modeling submissions for Chile's national selection process for the IMMC.	
RialStat	Santiago, Chile
Professional Intern	2023 – 2024
• Worked with economic and financial data, supporting forecasting and investment-portfolio models.	

LANGUAGES

Spanish: Native	English: C1 (IELTS Academic: 7.5 overall)
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